ELECTRONICS AND COMMUNICATION ENGINEERING

COMMUNICATION SYSTEM

Amplitude modulation

Lecture No.: 01

By- SAKET VERMA SIR

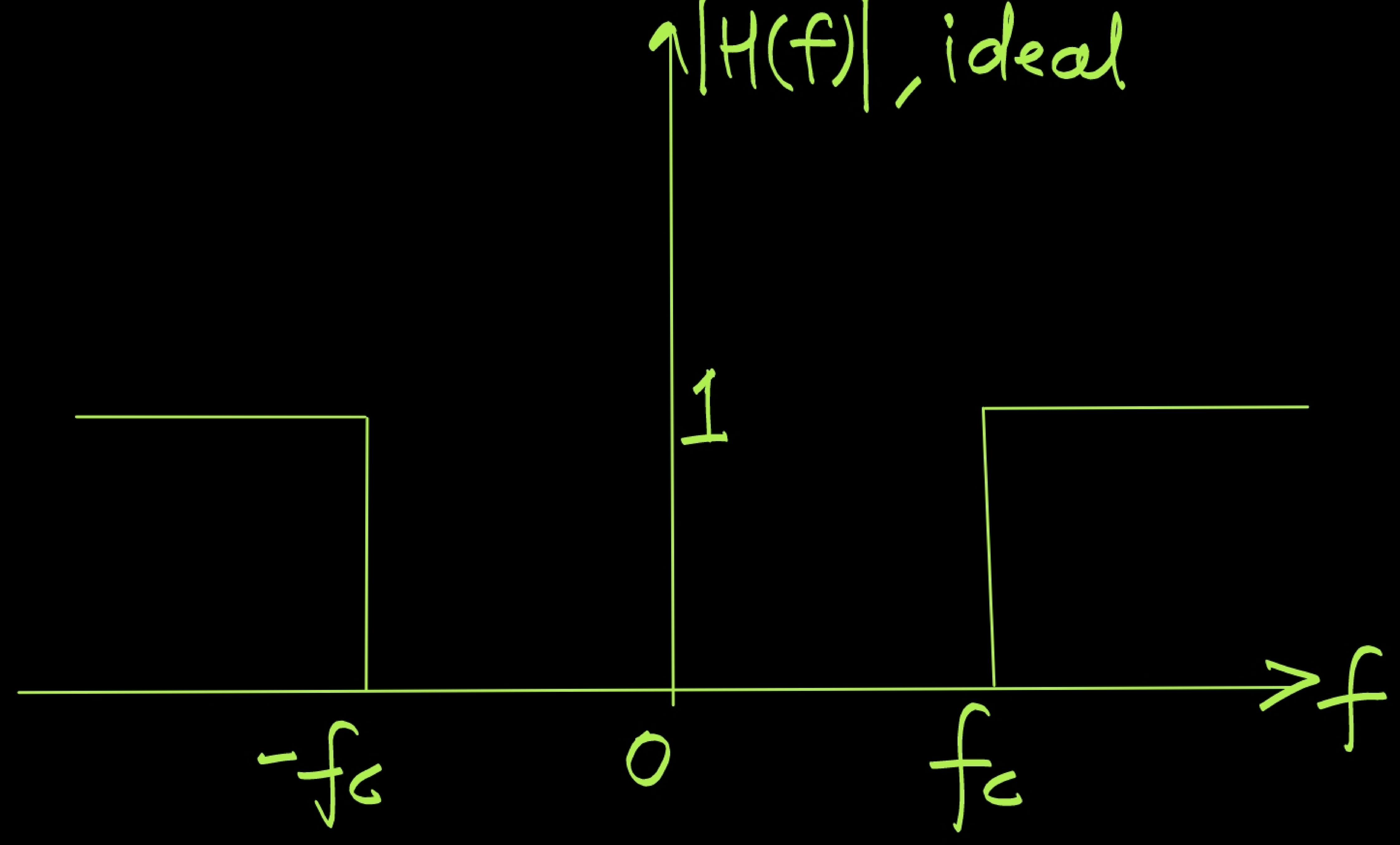
1. Fourier Series & Fourier Transform

1. Modulation
2. Filter

SAKET VERMA SIR

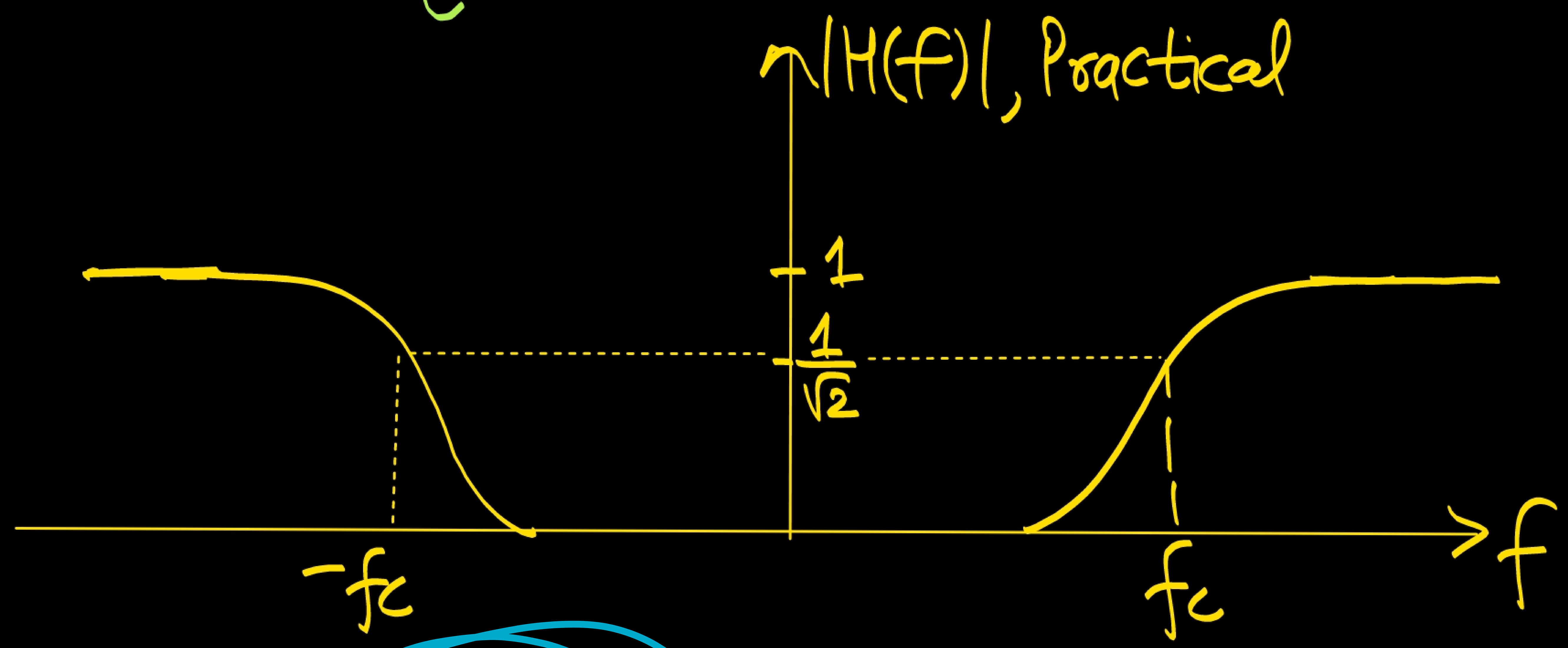
Filter

High Pass Filter:- A filter which passes frequencies higher than cut-off freq. is called HPF.



High pass filter

$$H(f) = \begin{cases} 1 ; & |f| \geq f_c \\ 0 ; & \text{Otherwise} \end{cases}$$

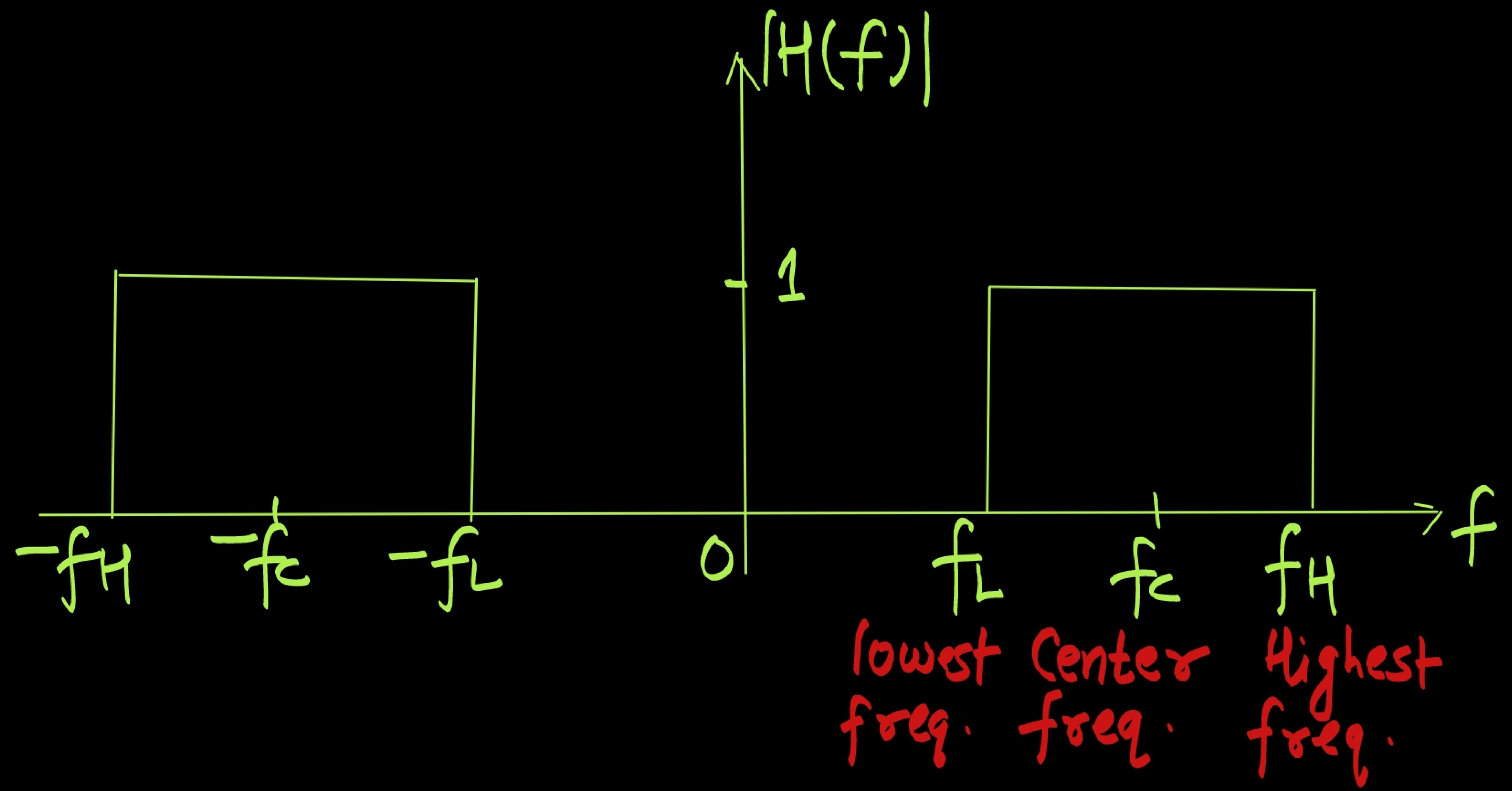


$1/\sqrt{2}$

Filter

Bandpass Filter

A filter which passes band of freq. is called BPF.



$$|H(f)| = \begin{cases} 1 ; & f_L \leq |f| \leq f_H \\ 0 ; & \text{otherwise} \end{cases}$$

$$H(f) = \text{rect}\left(\frac{f-f_c}{f_H-f_L}\right) + \text{rect}\left(\frac{f+f_c}{f_H-f_L}\right)$$

IFT

let $B = f_H - f_L$.

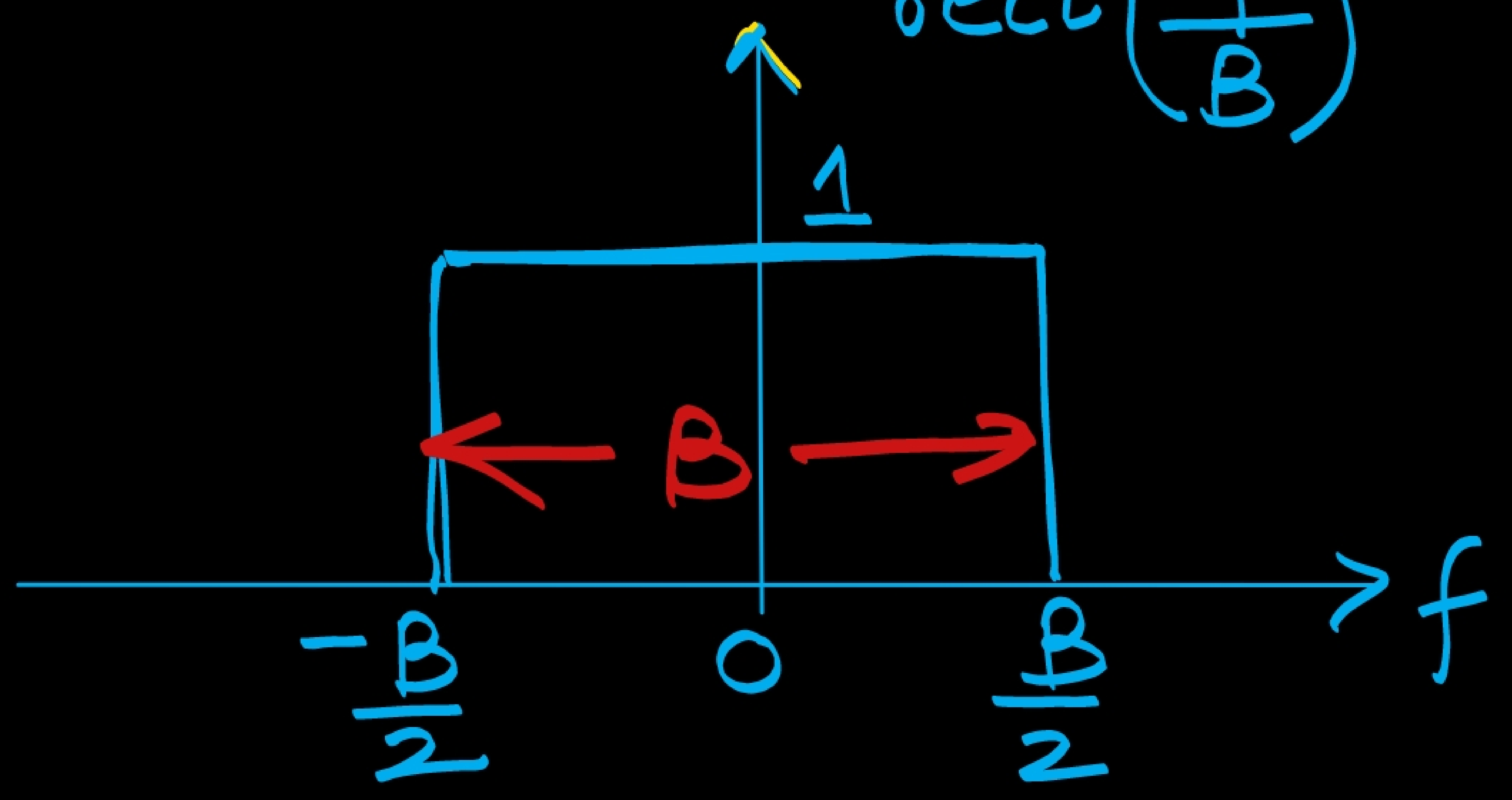
$$h(t) = 2B\,\text{sinc}(tB)\cos(2\pi f_c t)$$

$$H(f) = \text{rect}\left(\frac{f-f_c}{f_H-f_L}\right) + \text{rect}\left(\frac{f+f_c}{f_H-f_L}\right)$$

$$H(f) = rect\left(\frac{f-f_c}{B}\right) + rect\left(\frac{f+f_c}{B}\right)$$

$$AT\,sinc(tT) \xleftrightarrow{F.T.} \underbrace{A\,rect\left(\frac{f}{T}\right)}_{\text{Known}}$$

$rect\left(\frac{f}{B}\right)$



$$AT\,sinc(tT) \longleftrightarrow A\,rect\left(\frac{f}{T}\right)$$

$T = B$

Compare, $T = B$, $A = 1$

$$\underline{B\,sinc(tB)}$$

$$\boxed{2B\,sinc(tB)\cos\omega_c t \xleftrightarrow{F.T.} rect\left(\frac{f-f_c}{B}\right) + rect\left(\frac{f+f_c}{B}\right)}$$

Taking IFT $= B\,sinc(tB)$

$$e^{j2\pi f_c t} B\,sinc(tB) \longleftrightarrow rect\left(\frac{f-f_c}{B}\right)$$

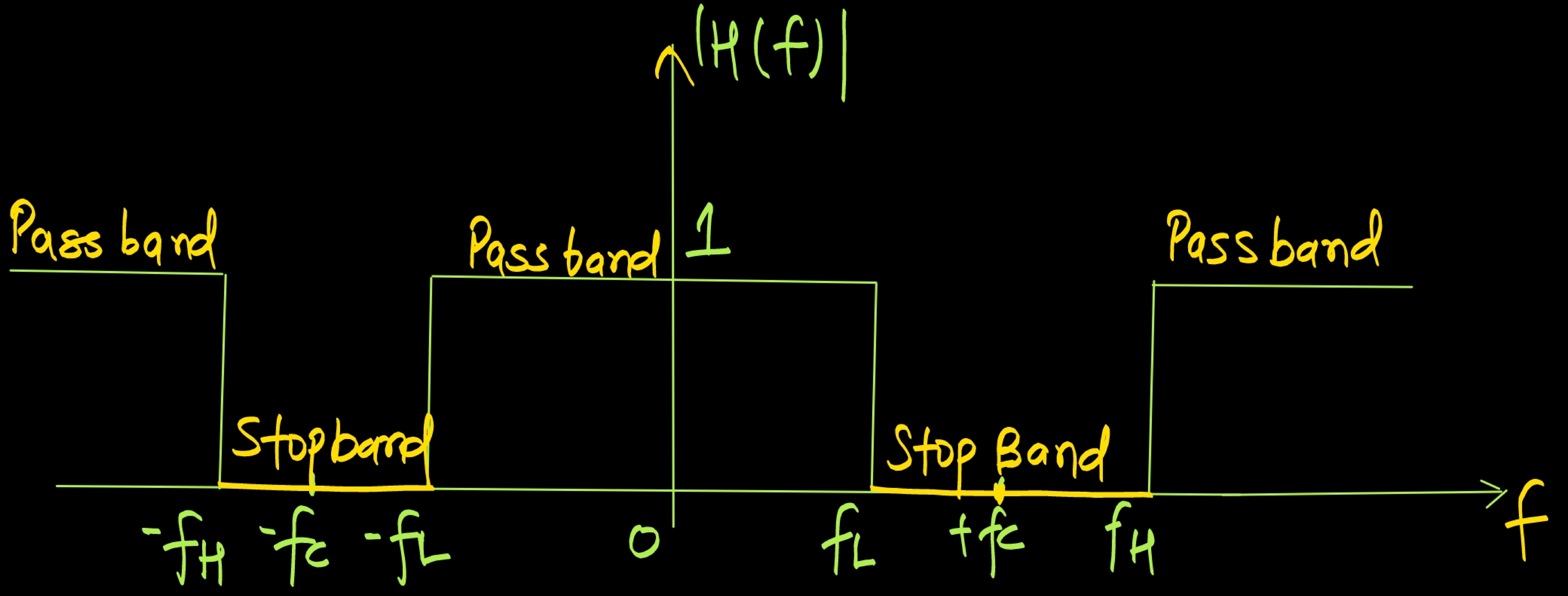
$$e^{-j2\pi f_c t} B\,sinc(tB) \longleftrightarrow rect\left(\frac{f+f_c}{B}\right)$$

$A = 1$ $\quad B\,sinc(tB)\cos(\omega_c t)$

$$AT\,sinc(tT) \xleftrightarrow{FT} A\,rect\left(\frac{f}{T}\right)$$

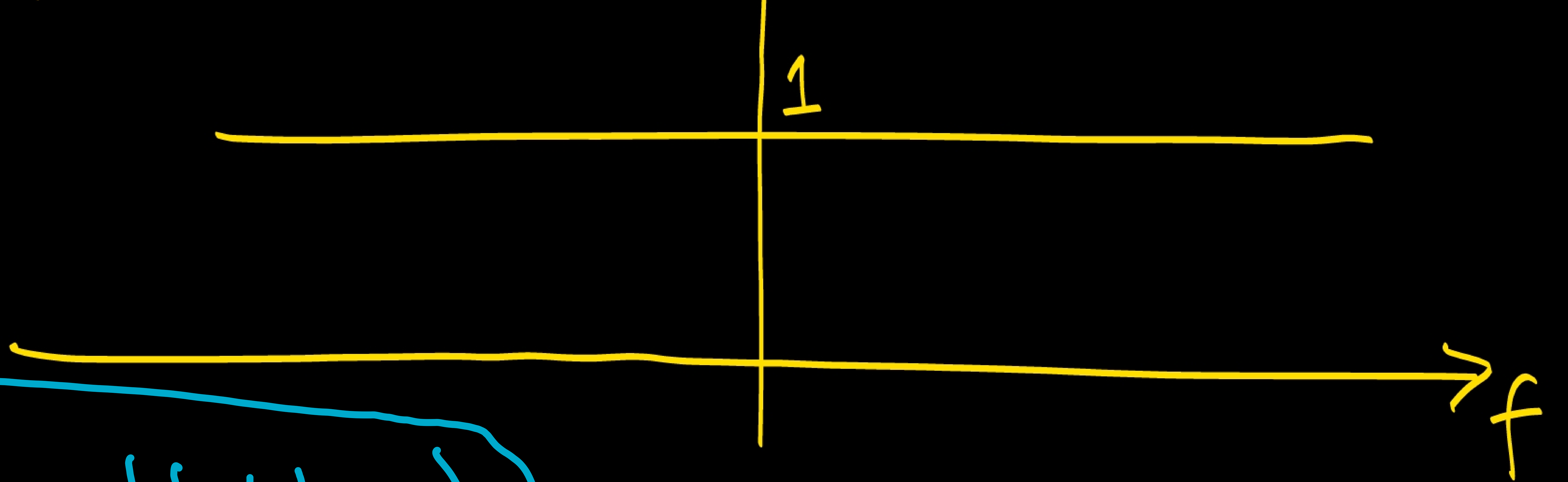
Band stop filter (Notch filter)

A filter which reject band of freq. is called Band reject/stop filter.



$$H(f) = \begin{cases} 0 \; ; & f_L \leq |f| \leq f_H \\ 1 \; ; & \text{otherwise} \end{cases}$$

All pass filter, $|H(f)| = 1 \;\; \forall f$



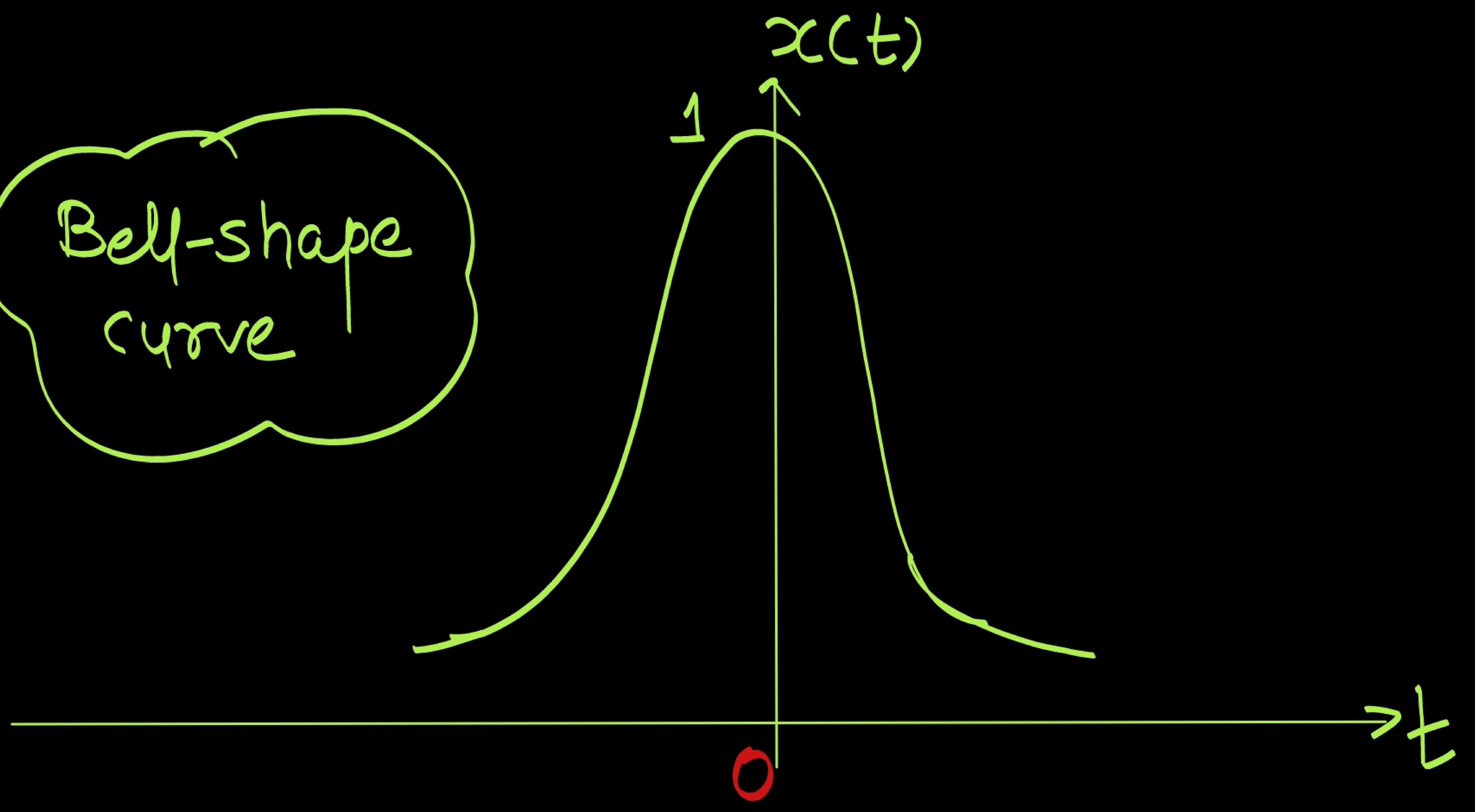
Band stop filter

$$\boxed{H(f) = 1 - 1\,\text{rect}\left(\frac{f-f_c}{f_H-f_L}\right) + \text{rect}\left(\frac{f+f_c}{f_H-f_L}\right)}$$

Gaussian Signal

$$x(t) = e^{-\pi t^2}$$

Bell-shape curve



Standard Fourier Transform

$A\,\mathrm{rect}\left(\frac{t}{T}\right) \longleftrightarrow AT\,\mathrm{sinc}(fT)$

Signal	Fourier Transform
$\delta(t)$	1
$A\,\mathrm{rect}\left(\frac{t}{T}\right)$	$AT\,\mathrm{sinc}(fT)$
$e^{-at}u(t)$	$\frac{1}{a+j\omega}$
$e^{-a\lvert t\rvert}$	$\frac{2a}{a^2+\omega^2}$
$u(t)$	$\frac{1}{j\omega}+\pi\delta(\omega)$

Signal	Fourier Transform
$A\,\mathrm{tri}\left(\frac{t}{T}\right)$	$AT\,\mathrm{sinc}^2(fT)$
$e^{-\pi t^2}$	$e^{-\pi f^2}$
1	$2\pi\delta(\omega)$
$\mathrm{sgn}(t)$	$\frac{2}{j\omega}$
$\cos\omega_0 t$	$\frac{1}{2}\left[\delta(f-f_0)+\delta(f+f_0)\right]$
$\sin\omega_0 t$	$\frac{1}{2j}\left[\delta(f-f_0)-\delta(f+f_0)\right]$

$\frac{1}{a+j\omega}$

$u(t) \longleftrightarrow \frac{1}{j\omega} + \pi\delta(\omega)$

Energy and Area

Signal	Area	Energy
$\text{sinc}(t)$	1	1
$Sa(t)$	π	π
$A\,\text{rect}\left(\frac{t}{T}\right)$	AT	A^2T
$A\,\text{tri}\left(\frac{t}{T}\right)$	$\frac{1}{2}\times 2A\times T$	$\frac{2A^2T}{3}$
$e^{-\pi t^2}$	1	$\frac{1}{\sqrt{2}}$

If $x(t) \xrightarrow{\text{Area}} A$

then $Kx(-\alpha t \pm t_0) \xrightarrow{\text{Area}} \left|\frac{KA}{\alpha}\right|$

if $x(t) \xrightarrow{\text{Energy}} E$

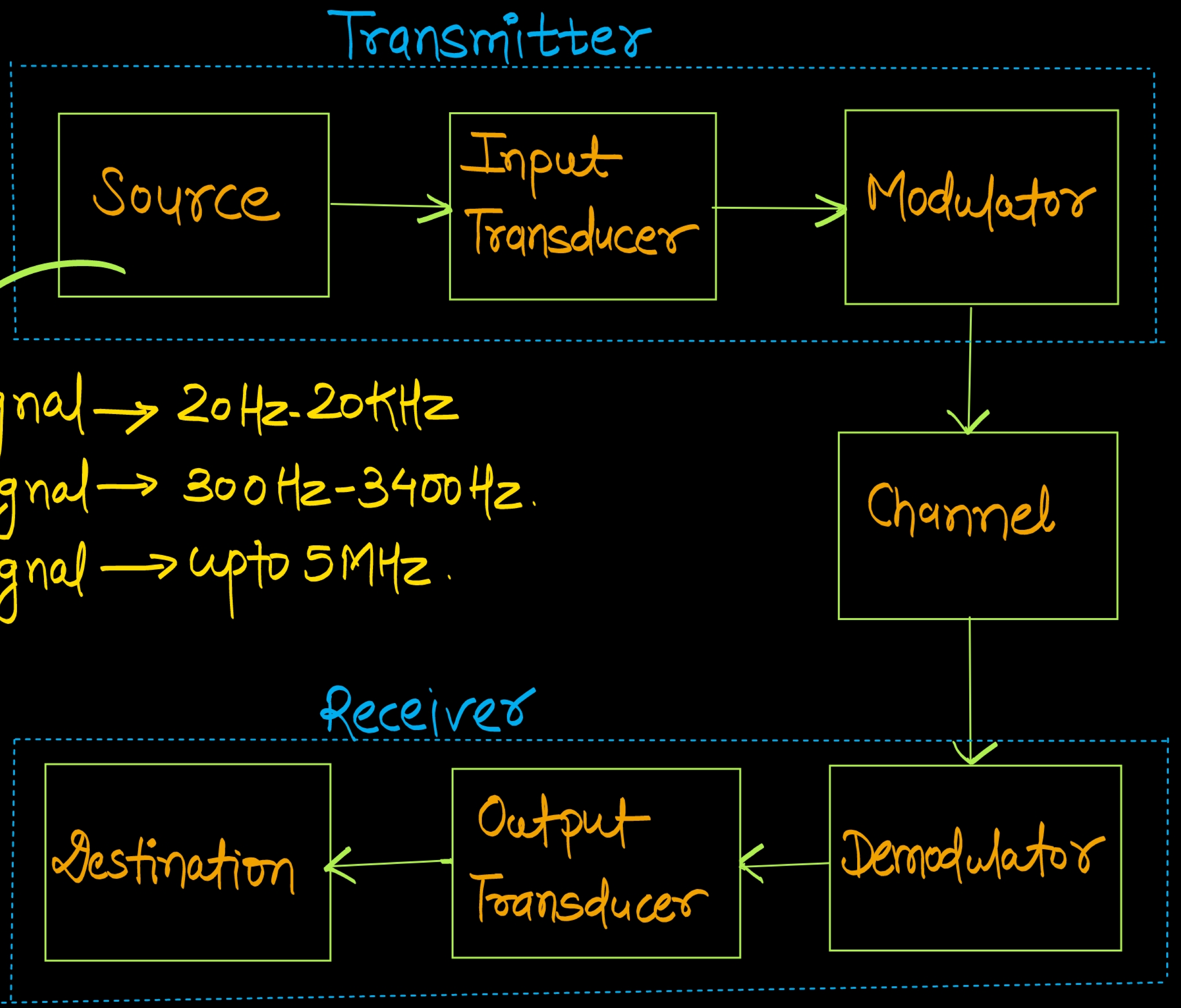
then $Kx(-\alpha t \pm t_0) \xrightarrow{\text{Energy}} \frac{K^2E}{\alpha}$

Energy and area

sinc (t) 1 1

S

Block Diagram of Communication System

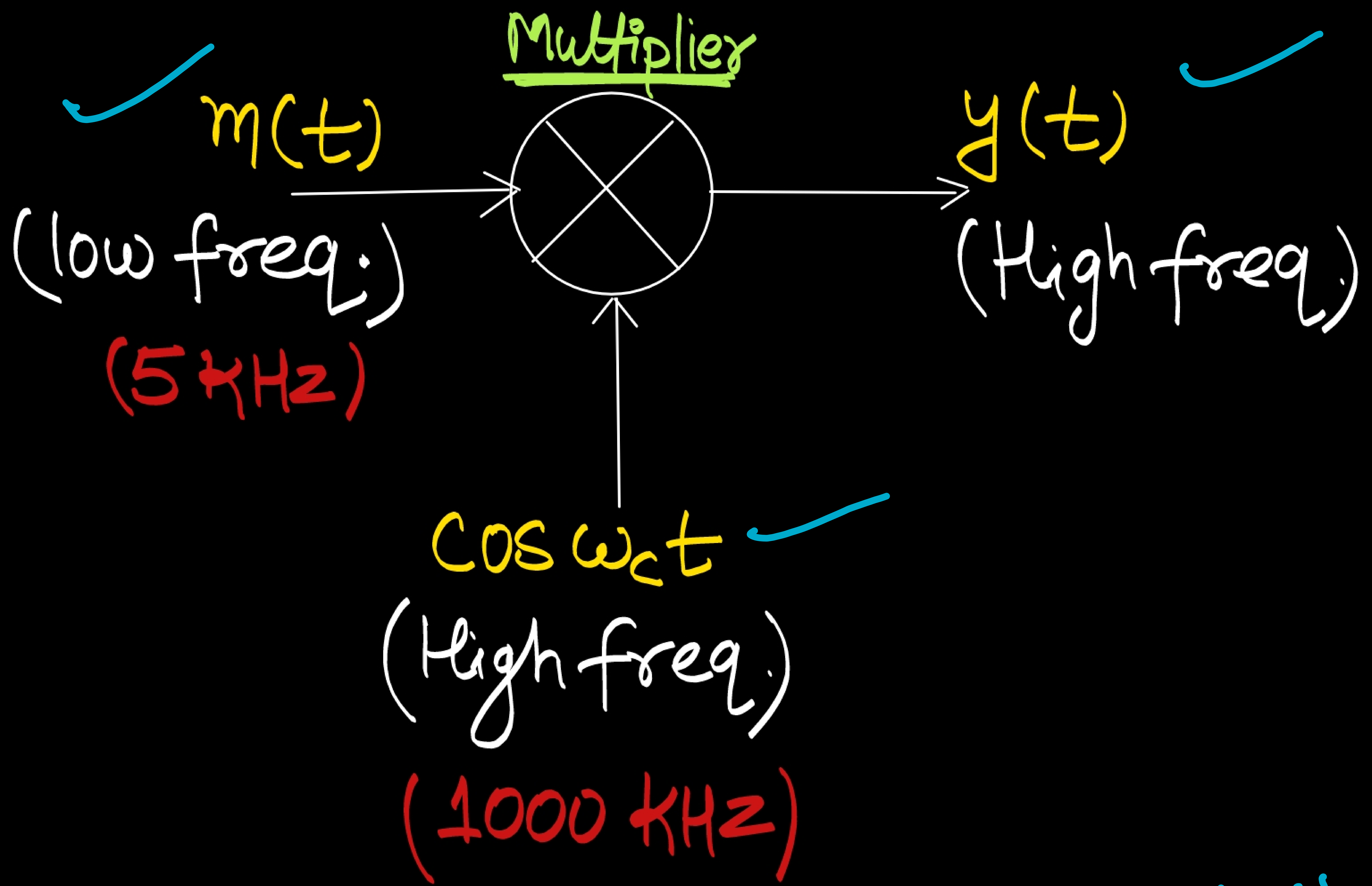


audio signal → 20 Hz - 20 KHz
voice signal → 300 Hz - 3400 Hz.
video signal → upto 5 MHz.

Modulation

- It is the process in which low frequency signal is translated into high frequency signal so also known as Frequency Translation.
- It is the process in which characteristics of one waveform (Carrier) is varied in accordance with characteristics of another waveform (Modulating signal or Message).

How to perform modulation?



$$\boxed{f_c >> f_m}$$

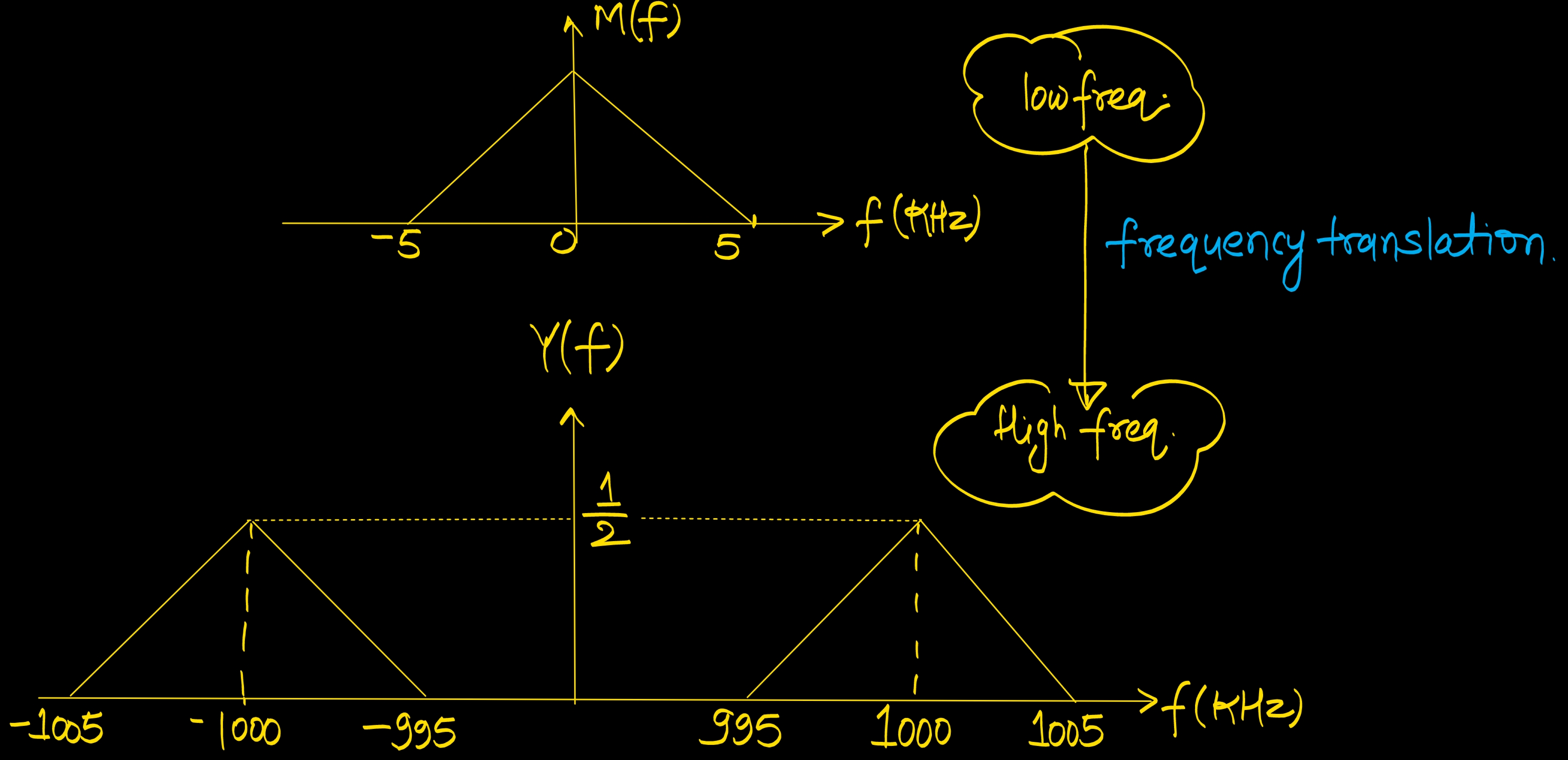
$$y(t) = m(t)\cos\omega_c t$$

Taking Fourier Transform.

$$Y(f) = \frac{1}{2}\left[M(f-f_c) + M(f+f_c)\right]$$

$$= \frac{1}{2}\left[M(f-1000) + M(f+1000)\right]$$

$$y(t) = m(t)\cos(\omega_c t)$$


M(f)
f (KHz)
-5
0
5
Y(f)
1/2
-1005
-1000
-995
995
1000
1005
f (KHz)
low freq.
frequency translation
high freq.

Thank you

GW

Soldiers !